William Edward Hahn, Ph.D.

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SUMMARY

AI researcher and educator specializing in compressed sensing, deep learning, computer vision, and computational neuroscience. Founded the Machine Perception and Cognitive Robotics Lab (20 projects, 14 researchers) and led construction of FAU's $1M Gruber AI Sandbox. 13+ peer-reviewed publications including *Nature: Scientific Reports*. Developed cross-departmental AI curricula for Biology, Geoscience, and Chemistry. College of Science Teacher of the Year (2023). Featured in *Science* Magazine and the *Theories of Everything* podcast.

EDUCATION

Ph.D., Complex Systems and Brain Sciences, Florida Atlantic University 2016
Graduate Study, Mathematics, University of North Carolina at Greensboro 2011
B.S., Mathematics and Physics, Guilford College 2008

EXPERIENCE

Associate Director, Center for Complex Systems, FAU 2025–Present
Associate Professor, Dept. of Mathematical Sciences, FAU 2024–Present
Assistant Professor, Dept. of Mathematical Sciences, FAU 2018–2024
Research Assistant Professor, Center for Complex Systems, FAU 2016–2018

RESEARCH LEADERSHIP

MPCR Lab — Co-founded 2014. 20 active projects, 14 researchers (Spring 2026). Eight focus areas: LLMs/VLA models, artificial life, unconventional computing, swarm intelligence, AI safety, computer vision, cognitive science, edge computing. Featured in *Science* Magazine (2019).

Gruber AI Sandbox — Led construction of $1M, 3,400 sq ft multi-disciplinary AI lab in FAU's Wimberly Library. 10 GPU workstations, VR, collaborative spaces. Hosts Mindfest, IEEE Robotics, AI lecture series. Speakers have included Stephen Wolfram, Yoshua Bengio, David Chalmers, Daniel Dennett, Eliezer Yudkowsky.

SELECTED PUBLICATIONS

- Jerez et al. (2021). Deep learning immunogold particle identification. *Nature: Scientific Reports*, 11, 7771.
- Teti, **Hahn** et al. (2021). Behaviorally-cloned DNNs in autonomous steering. *Robotics and Autonomous Systems*, 142.
- Morris et al. (2020). Transformer embeddings for binding prediction. *J. Chemical Information and Modeling*, 60(9).
- Sachdeva et al. (2020). GANs for robotic telesurgery latency. *J. Robotic Surgery*, 15.
- Barenholtz et al. (2021). NPS surveillance via Reddit + ML. *Int. J. Drug Policy*, 98.
- **Hahn** et al. (2015). Deep learning human actions via sparse filtering. *Multimedia Tools and Applications*, 74.

11 journal articles, 2 conference proceedings, 1 preprint. Full list: Google Scholar.

FUNDING HIGHLIGHTS

- Gruber Foundation: $1,000,000 (Sandbox, 2019) + $330,000 (Lab, 2017–2018) + $210,000 (VLA research + annual, 2024–2026)
- U.S. Air Force SBIR Phase II (PI): $1,249,791 (2025–2026)
- Ghost Robotics: $62,000 (2018) · Nvidia: $12,000 (2018) · FAU Seed: $16,000 (2018)

TEACHING

Courses developed: AI for Science (Spring 2026) · AI in Medical Image Analysis · Mathematics of Data Science · Fractal Geometry · AI in Science and Medicine · Max Planck Data Science

Cross-departmental AI modules: Built 6-notebook zero-to-AI curricula (PyTorch, Google Colab) deployed in Biology (birdsong classification, with Prof. Anderson), Geoscience (NOAA climate data, with Prof. Johanson), and Chemistry (molecular analysis, with Prof. Lepore). Biology module in its 2nd semester.

Additional: 100+ open-source GitHub repositories for AI education. ChatGPT integration in course materials. AI Teaching Workshop for College of Science faculty.

STUDENT OUTCOMES (SELECTED)

- Los Alamos National Lab (2 graduates)
- Harvard Law School
- Northwestern Ph.D. (EE)
- Naval Postgraduate School
- U of Tampa (tenure-track)
- Brown University (Neuroscience)
- Wolfram Summer School
- Founded Simuli.ai (AI startup)
- NSF Fellowship (M.S. in AI)
- Notre Dame Post-doc

SELECTED TALKS AND MEDIA

- *Theories of Everything* podcast — 4 episodes (2024–2025)
- Wolfram Institute livestream (2025)
- MIT Media Lab hackathon (2025) · Beth Israel Hospital AI Center (2025)
- *Science* Magazine: "Could robots be psychology's new lab rats?" (2019)
- Keynote: "From Turing to Hyperorganisms," Frontier Tower, San Francisco (2025)
- NIU AI Safety Conference, Washington, DC (2023)
- Lab coverage: BBC, Reuters, CBS, Fox, Scientific American, Vox, Wired

HONORS

Teacher of the Year, FAU College of Science (2023) · Multiple First Place awards at FAU Wave, GPSA, and Broward County Science Fairs · NASA Special Recognition Award (2018)